

INPUT

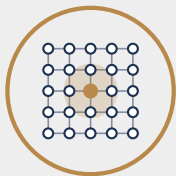


Multimodal data

`data/general_main.csv`

138 measurements, 8 features:
nanomechanics, Raman, fracture

LEARN



Train the SOM

`som-train`

SOMPY map, PCA initialization;
codebook saved as HDF5

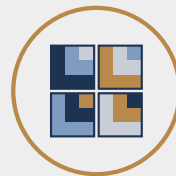
INTERPRET



Cluster the
codebook

`som-analysis`

k-means on the SOM codebook;
 k from elbow and silhouette



Interpretable
maps

`som-visualize`

component planes, U-matrix,
cluster maps by metadata

`pip install som-multimodal-data-reduction` · packaged analysis of Renteria *et al.*, JMBBM 129, 105147 (2022) · runs on CPU